General introduction of working mode

For X1/X3-ESS G4

Storage&Battery

ENGLISH

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Introduction

The G4 energy storage inverter has 7 working modes and two sets of flexible time axes. Except for EPS, the inverter automatically enters according to the working conditions, and other modes need to be manually selected by the customer.
Working mode: Self Use, Feed-in priority, Backup mode, EPS, Manual, Generator mode, peak shaving.
time axis:Allowed discharging period、forced charging period.
The three modes of Self Use, Feed-in priority, and Backup mode can be combined with two sets of timelines.
Generator mode, peak shaving need refer to additional KB:

1.Peak shaving

2.Generator mode

Working modes

Self Use (Priority: Loads ＞ Battery ＞ Grid):
The self-use mode is suitable for areas with low feed-in subsidies and high electricity prices.
The power of PV will supply the loads first, and surplus power will charge the battery, then the remaining power will feed into the grid.

- Min SoC: (Default: 10% range: 10%~100%)

The minimum SOC of the system, that is, the battery will not discharge when the SOC of the battery reaches this value.

- Charge from grid:

Users can set whether to allow battery Charge from grid, if this function is enabled, the inverter will take power to charge the battery during the forced charging period.

- Charge battery to: (Default: 30% range: 10%~100%)

Customers can set their own target value, i.e. during the forced charging period, the inverter will use both PV & GRID energy to charge the battery SOC to the target SOC value +5% as much as possible, after the battery SOC meets the target value, if the PV energy is still sufficient, the inverter will continue to use PV energy to charge the battery.

Feed-in priority(Priority: Loads ＞ Grid ＞ Battery):
The feed-in priority mode is suitable for areas with high feed-in subsidies, but has feed-in power limitation.
The power of PV will supply the loads first, and surplus power will feed into the grid, then the remaining power will charge the battery.

- Min SoC: (Default: 10% range:10%~100%)

The minimum SOC of the system, that is, the battery will not discharge when the

SOC of the battery reaches this value.

- <u>Charge battery to: (Default: 100% range: 10%~100%)</u>

Customers can set their own target value, i.e. during the forced charging period, the inverter will use both PV & GRID energy to charge the battery SOC to the target SOC value +5% as much as possible, after the battery SOC meets the target value, if the PV energy is still sufficient, the inverter will continue to use PV energy to charge the battery.

Backup mode (Priority: Loads ＞ Battery ＞ Grid):

The back-up mode is suitable for areas with frequent power outages.

This mode will maintain the battery capacity at a relatively high level, to ensure that the emergency loads can be used when the grid is off.

Similar to the working logic of "self-use" mode, the biggest difference is that the inverter will enter Idle mode in self-use mode without PV energy & battery SOC=Min SOC, and the inverter will enter standby in backup mode to deal with unexpected situations such as sudden power outages

- <u>Min SoC: (Default: 30% range:10%~100%)</u>

The minimum SOC of the system, that is, the battery will not discharge when the SOC of the battery reaches this value.

- <u>Charge battery to: (Default: 50% range:30%~100%)</u>

In this mode, the charge form grid function is turned on by default, and customers can set the target value by themselves, that is, during the forced charging period, the inverter will cooperate with PV&GRID to charge the battery to the target value as much as possible.

In the above three working modes (Self Use, Feed-in priority, Backup mode), if the customer needs to set 0 export, the user can set the value of the export control function to "0", which means that the system will not discharge to the grid ,

Manual

This work mode is for the after-sales team to do after-sales maintenance.

Customers can set forced charging, forced discharging, and stop charging and discharging.

- <u>Forced Discharge</u>

Force the system to discharge

- <u>Forced Charge</u>

Force the system to charge

- <u>Stop Chrg&Dischrg</u>

Stop system charging and discharging

EPS (Off-grid) (Priority: Loads ＞ Battery) :

In case of power failure, the system will supply EPS loads through PV and battery. (The battery must be installed, and EPS loads shall not exceed battery's max. output power.)

The power of PV will charge the loads first, and surplus power will charge the battery.

Forced charging period (Default value 00:00~00:00) :
The priority of forced charging period is higher than all work modes. Under the forced charging period, the inverter will charge the battery first until the battery SOC reaches the value of "charge battery to".

Allowed discharging period (Default value 00:00~23:59) :
Under the allowed discharging period, the inverter will allow the battery to discharge (but not force the battery to discharge). The following work modes will take effect under the allowed discharging period.

- Chrg&Dischrg Period

  Charge Period (Start Time/ End Time)

  Allowed Disc Period (start Time/ End Time)

Customers can set the charge and discharge time axis according to their own needs. The default time axis of the system is 24h to allow discharge

- Chrg&Dischrg Period2

  Charge Period(Start Time/ End Time)

  Allowed Disc Period(start Time/ End Time)

The second time axis is closed by default, and customers can open and set the charging and discharging time axis according to actual needs.

Typical scenario:

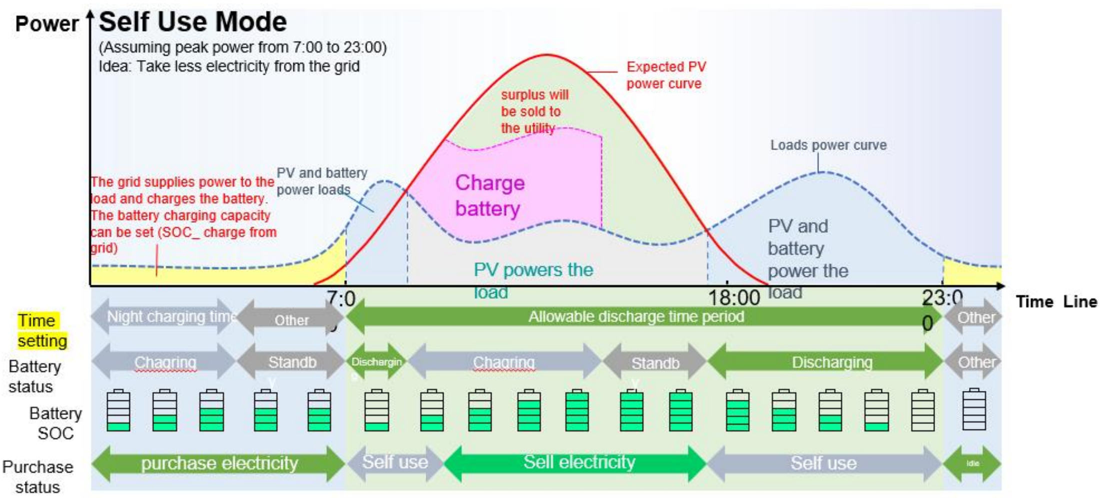
The electricity price in the customer area is very low from 2.00 to 4.00 in the morning, and the end user hopes to charge the battery to 100% during the time when the electricity price is cheap, so as to cope with the peak power consumption in the next morning.

- Chrg&DischrgPeriod

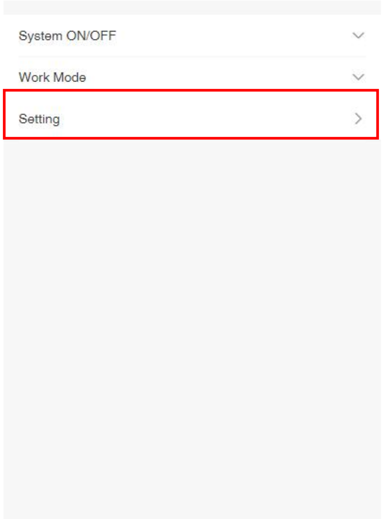
  Charge Period(Start Time/ End Time) (Default value 00:00~00:00)

  Allowed Disc Period(start Time/ End Time) (Default value 00:00~23:59)

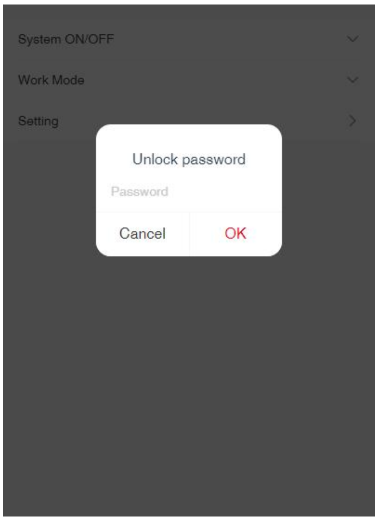
Adjust the Forced charging time axis from 00:00~00:00 to 2:00~4:00



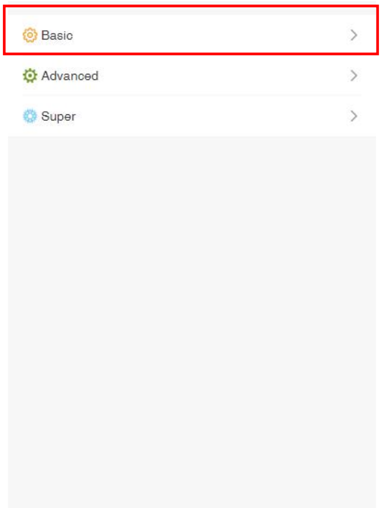
Setting steps:
1. Select and enter the setting interface



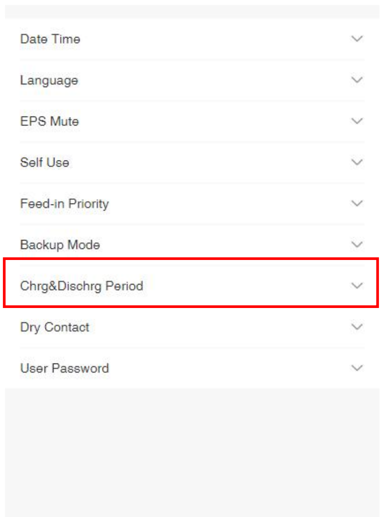
2.Enter the device password



3.Click "Basic"

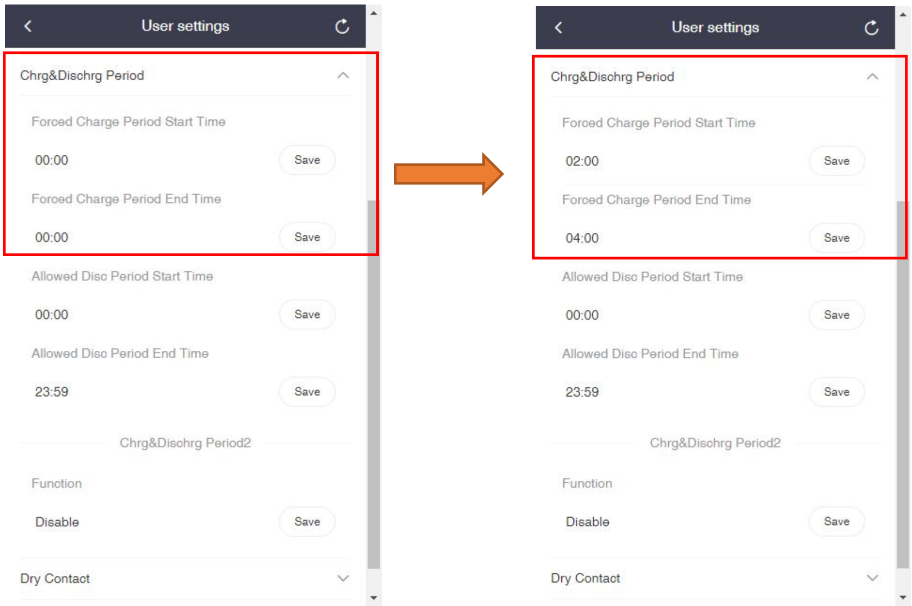


4. Click "Chrg&Dischrg period"



5. Adjust

Forced Charge Period Start Time from 0 to 2
Forced Charge Period End Time from 0 to 4



Compatibility

Inverter Type	Model	Work Mode
Hybrid Inverter	X1-Hybrid G4	Available
	X3-Hybrid G4	Available
	X3-HYB-G4 Pro	Available
	X1-IES	Available
	X3-IES	Available
	X1-Vast	Available
	X3-Ultra	Available